

On my farm I am using a robot to help me to collect strawberries. Majority of robot's collections are strawberries (super good; BIG and sweet); some are blueberries (bad; SMALL and sour).



You can download the data from course website.

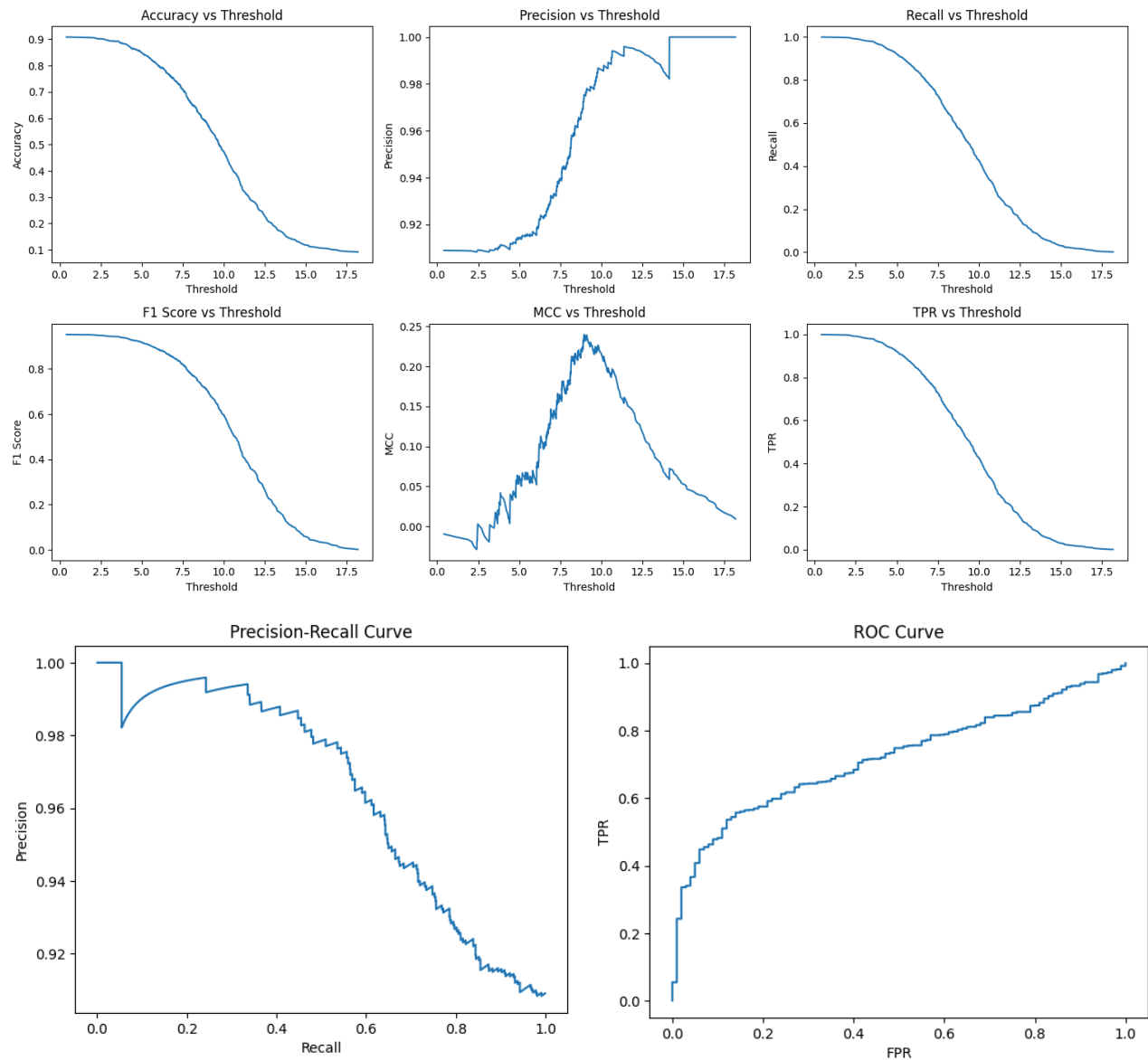
First row is the diameter of each individual fruit, second row is whether it is a strawberry or not.

Now I want to use a machine (by shaking with a net) to separate the strawberries and blueberries. Can you evaluate the performance difference between different size of the net? Only the diameter of the fruits matter to my machine.

Please draw the **Accuracy-Curves / Precision-Curves / Recall-Curves / F1-Score-Curves** for all different diameters.

Please draw the **PR-Curves** and the **ROC-Curves**

Please calculate the **AUC(Area under the curve)** of the ROC-Curve.



AUC Score = 0.7190